Applied Algorithms
Lecture 8: Route planning

October 22, 2025

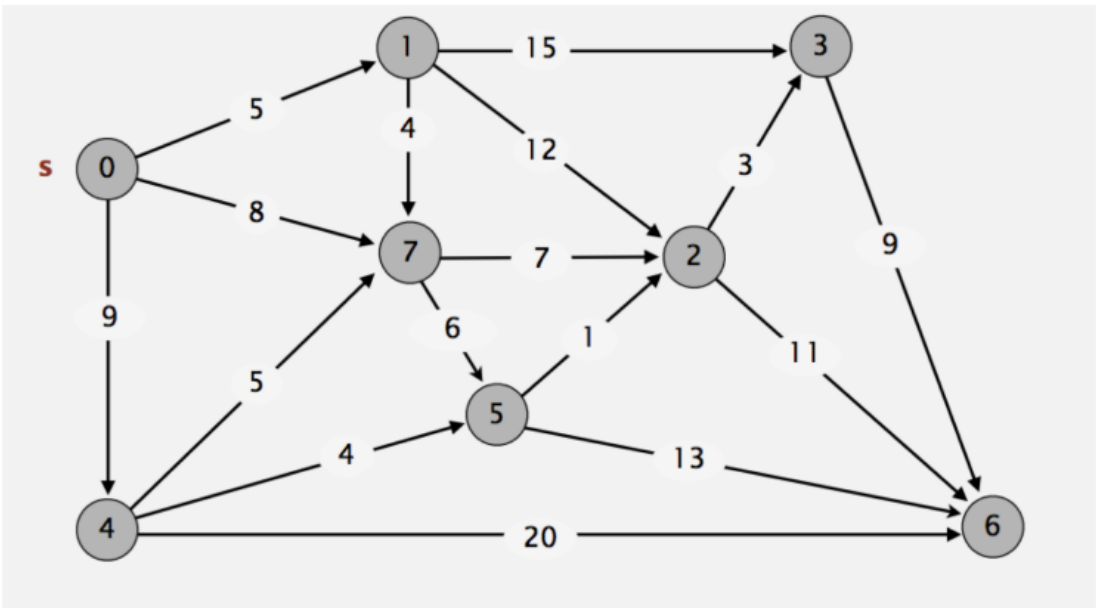
Martin Aumüller

Topic for today



Problem Single-Source-Shortest Paths

- **Input**: A directed, weighted graph $G = (V, E, c)$ with weight function $c: E \to \mathbb{R}_{\geq 0}$.
- **Task**: For given $s \in V$, compute (the length of) the shortest paths to every $v \in V$. We denote the length of a shortest path between $s, v \in V$ by $d(s, v)$.
- **Example**:
  - $d(s, 5) = 13$
  - $d(s, 6) = 25$



Dijkstra's Algorithm

- Data structure for graph with vertices $\{1, \ldots, n\}$:
  - Arrays distTo[1..n] and edgeTo[1..n]
  - distTo[v] is length of a shortest path from source to v
  - edgeTo[v] is vertex of the last edge on a shortest path from source to v
- Basic operation: *relax*(v, w)
  - If distTo[w] > distTo[v] + $c(v, w)$

    Set distTo[w] = distTo[v] + $c(v, w)$
    edgeTo[w]=v

High-level description of Dijkstra's algorithm

1. Consider vertices in increasing order of distance from s (by looking at the distTo-value)
2. Add vertex to tree, relax all outgoing edges from that vertex

Demo

Implementation considerations (n vertices, m edges)

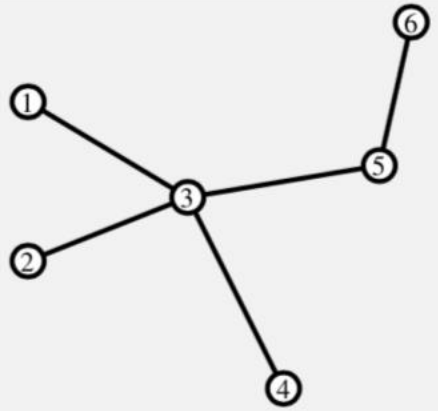
- **Question**: How to find the next node in the shortest path tree?
- **Solution 1**: Just search for smallest value in distTo array (as in demo)
  - Running time: $O(n^2)$
- **Solution 2**: Keep each node with its distTo value in a priority queue
  - Running time: $O(m \log n)$
  - ”eager”: each node is in the PQ at most once
  - PQ must support decreasing the priority of a key (Java’s PQ doesn’t)
- **Solution 3**: As solution 2, but ”lazy” variant
  - do not decrease, blindly insert
  - Running time: $O(m \log m) = O(m \log n)$ since $m \leq n^2$.

No difference asymptotically!

Where does the graph come from?

- Most popular source is openstreetmaps
- Format is XML
- Edge weights:
  - Distance between (lon, lat) pairs
  - Can take speed into consideration

```
<way>
    <nd ref="1">
    <nd ref="3">
    <nd ref="4">
</way>
<way>
    <nd ref="2">
    <nd ref="3">
    <nd ref="5">
    <nd ref="6">
</way>
```



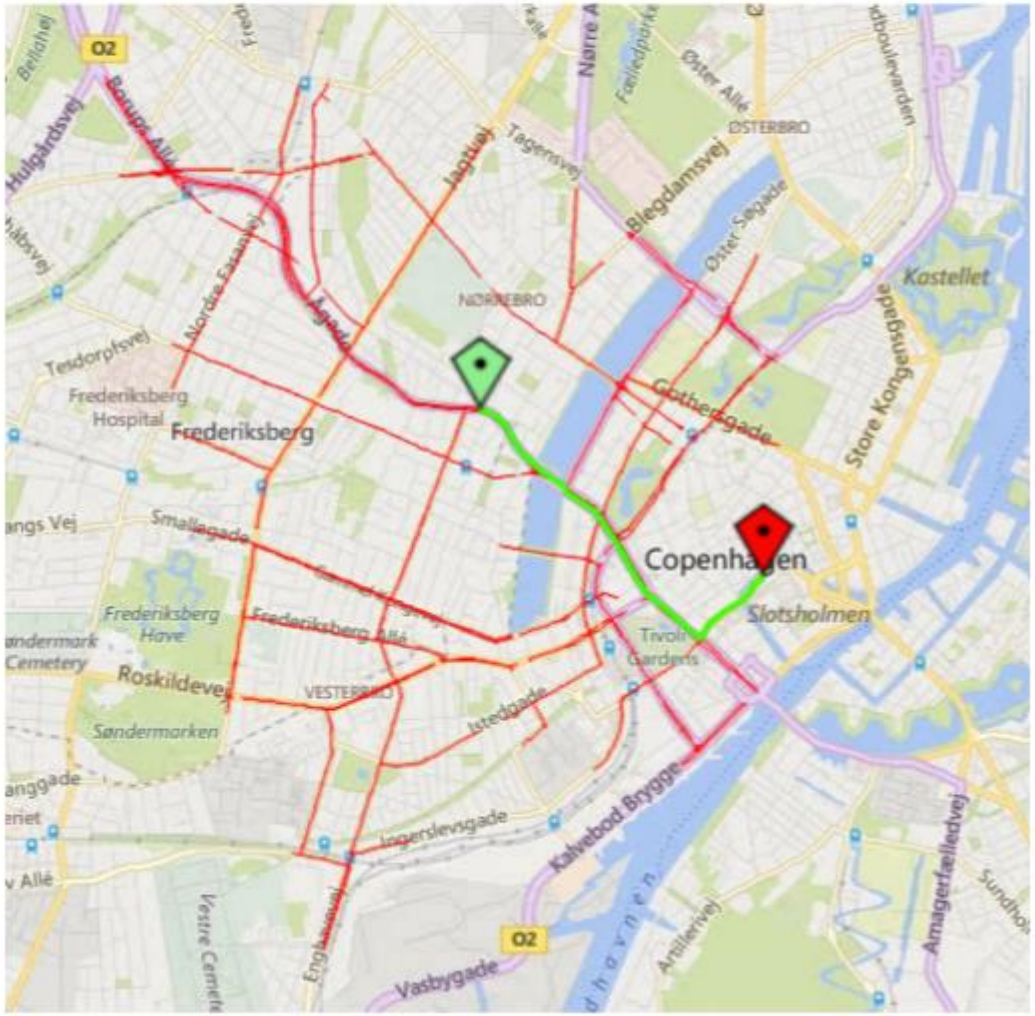


Denmark road network*: 570k vertices, 587k edges

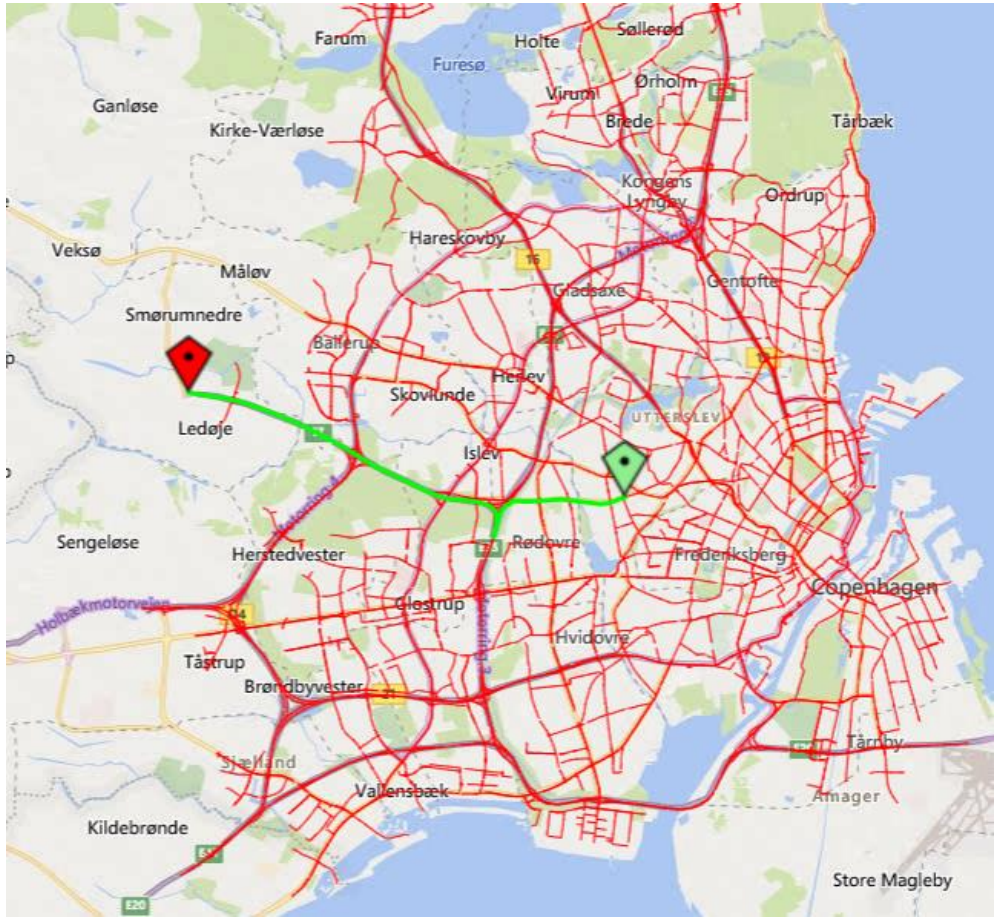
(* Actually much larger, we did some preprocessing.)

Improving Dijkstra

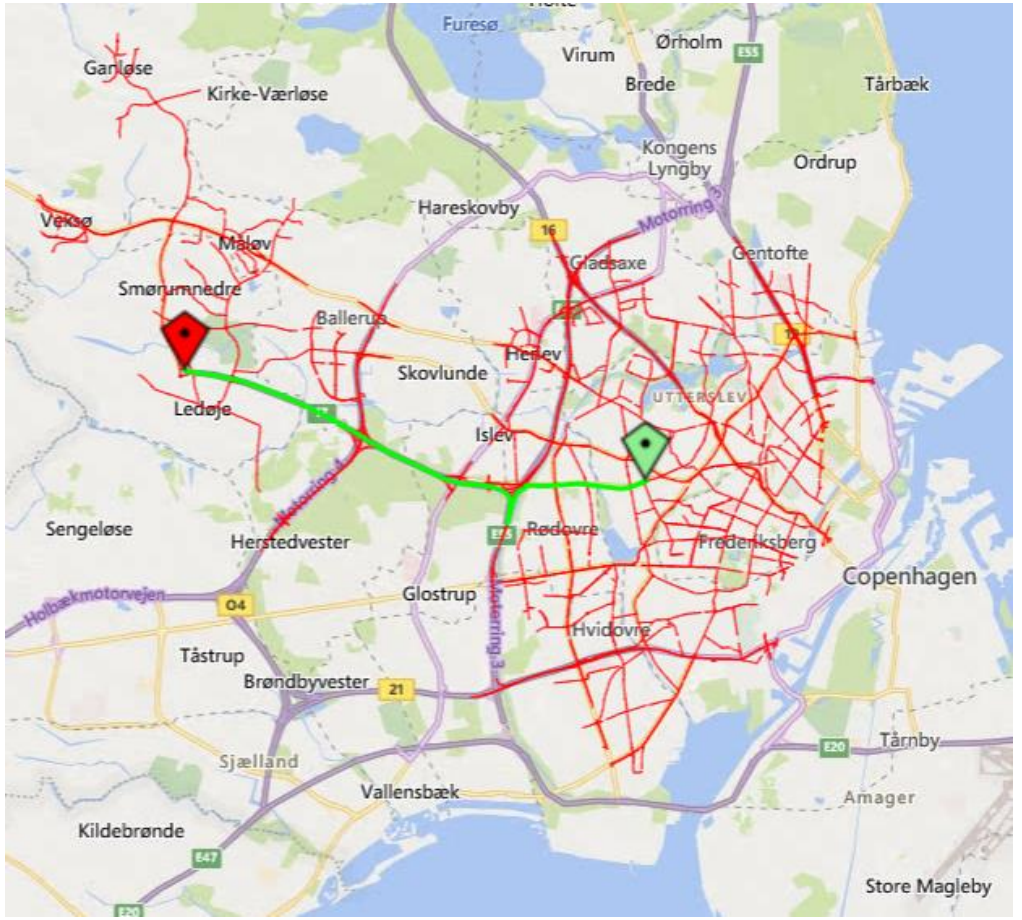
- Red: The shortest path tree
- Looks like a ”circle”, no orientation towards the goal



Bidirectional Dijkstra



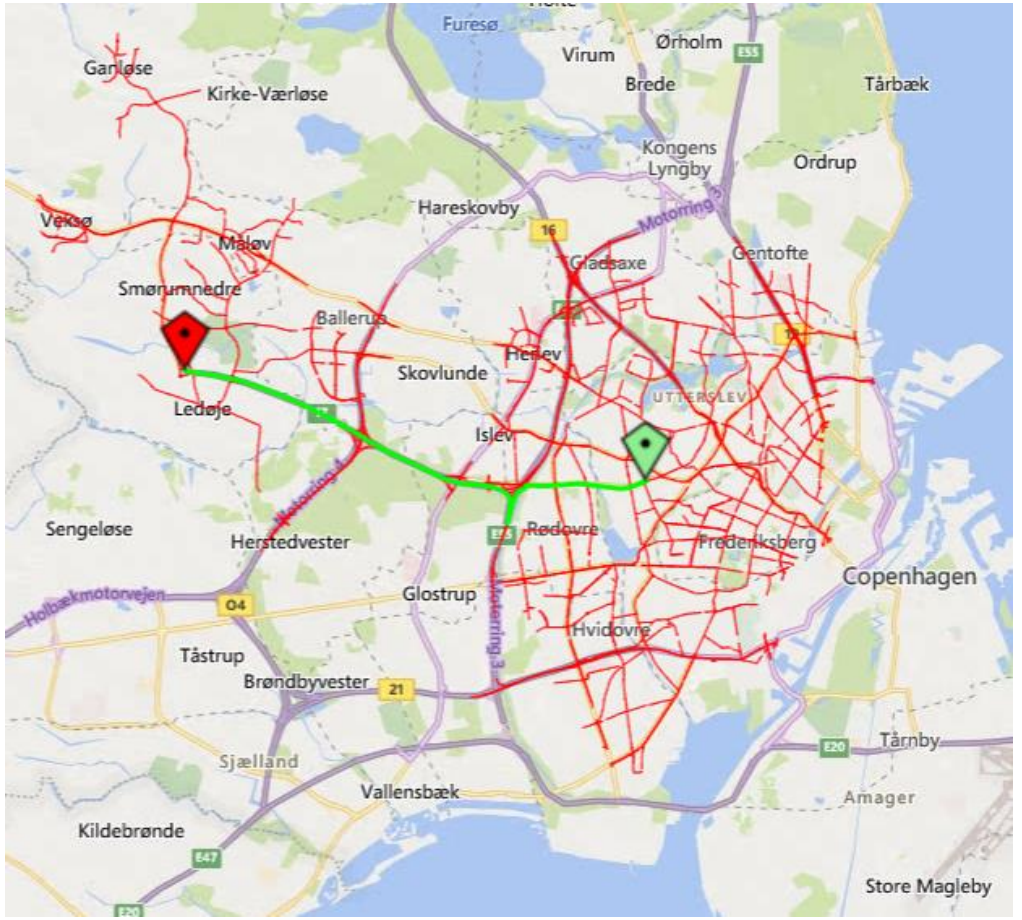
32713 edges in tree



13019 edges in tree

Outline of Bidirectional Dijkstra

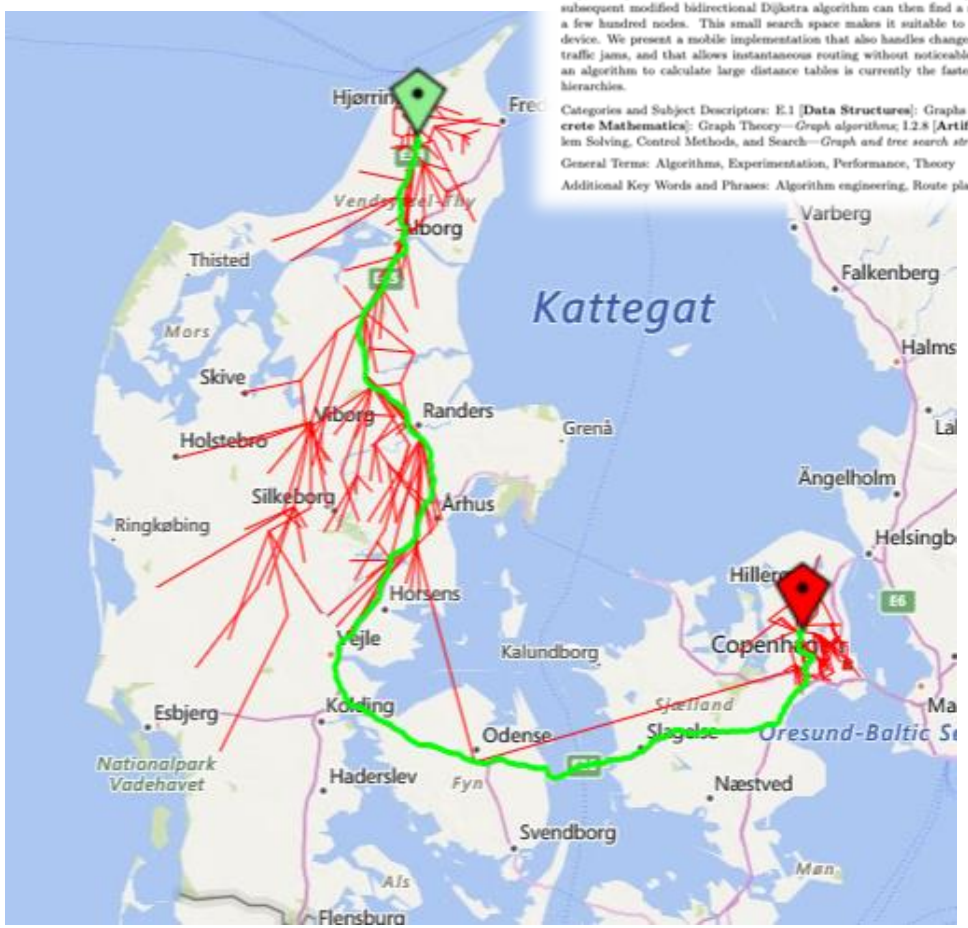
- Run Dijkstra both from source (forwards) and target (backwards)
- Always consider the node at minimum distance in both runs
  - Do not alternate
- When do you stop?
  1. Both runs consider same node?
  2. The current best distance between s, t is not longer than distances of next nodes in both priority queues summed up?
- Always faster?



Efficient Route Planning

- Status: Dijkstra + modifications
  - Hundreds of milliseconds query time on map of Denmark
- But what about larger maps?
- Or many users accessing your service at the same time?

- Many ideas where floating around in the past, **Contraction Hierarchies** by [Geisberger et al., 2008] is a particularly beautiful solution



316 edges in tree

Exact Routing in Large Road Networks using Contraction Hierarchies

ROBERT GEISBERGER, PETER SANDERS, DOMINIK SCHULTES and CHRISTIAN VETTER
Karlsruhe Institute of Technology, 76128 Karlsruhe, Germany

Contraction hierarchies are a simple approach for routing in road networks. Our algorithm calculates exact shortest paths and handles road networks of whole continents. During a preprocessing step, we exploit the inherent hierarchical structure of road networks by adding shortcut edges. A subsequent modified bidirectional Dijkstra algorithm can then find a shortest path visiting only a few hundred nodes. This small search space makes it suitable to implement it on a mobile device. We present a mobile implementation that also handles changes in the road network, like traffic jams, and that allows instantaneous routing without noticeable delay for the user. Also, an algorithm to calculate large distance tables is currently the fastest if based on contraction hierarchies.

Categories and Subject Descriptors: E.1 [**Data Structures**]: Graphs and networks; G.2.2 [**Discrete Mathematics**]: Graph Theory—*Graph algorithms*; I.2.8 [**Artificial Intelligence**]: Problem Solving, Control Methods, and Search—*Graph and tree search strategies*

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Outlook

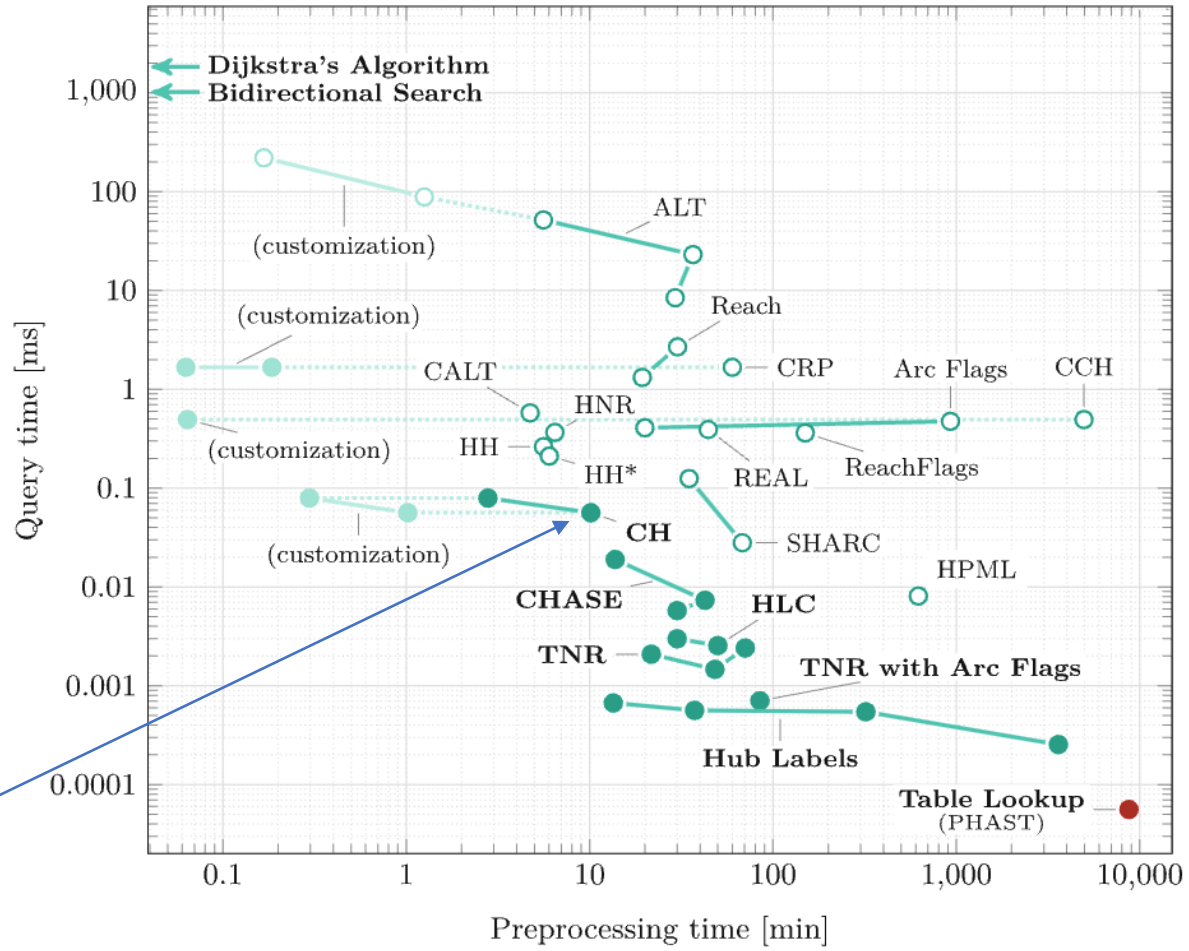


Fig. 7. Preprocessing and average query time performance for algorithms with available experimental data on the road network of Western Europe, using travel times as edge weights. Connecting lines indicate different trade-offs for the same algorithm. The figure is inspired by [238].

OUR FOCUS

[Bast et al., Route Planning in Transportation Networks, 2016]

Contraction Hierarchies Overview

- From input graph, construct *augmented graph*
  - Every node has a **rank**
  - New edges are created as **shortcuts**
- Augmented graph is constructed by *iteratively removing vertices* and adding edges such that *shortest paths are preserved*
- Bidirectional Dijkstra Search to find shortest paths

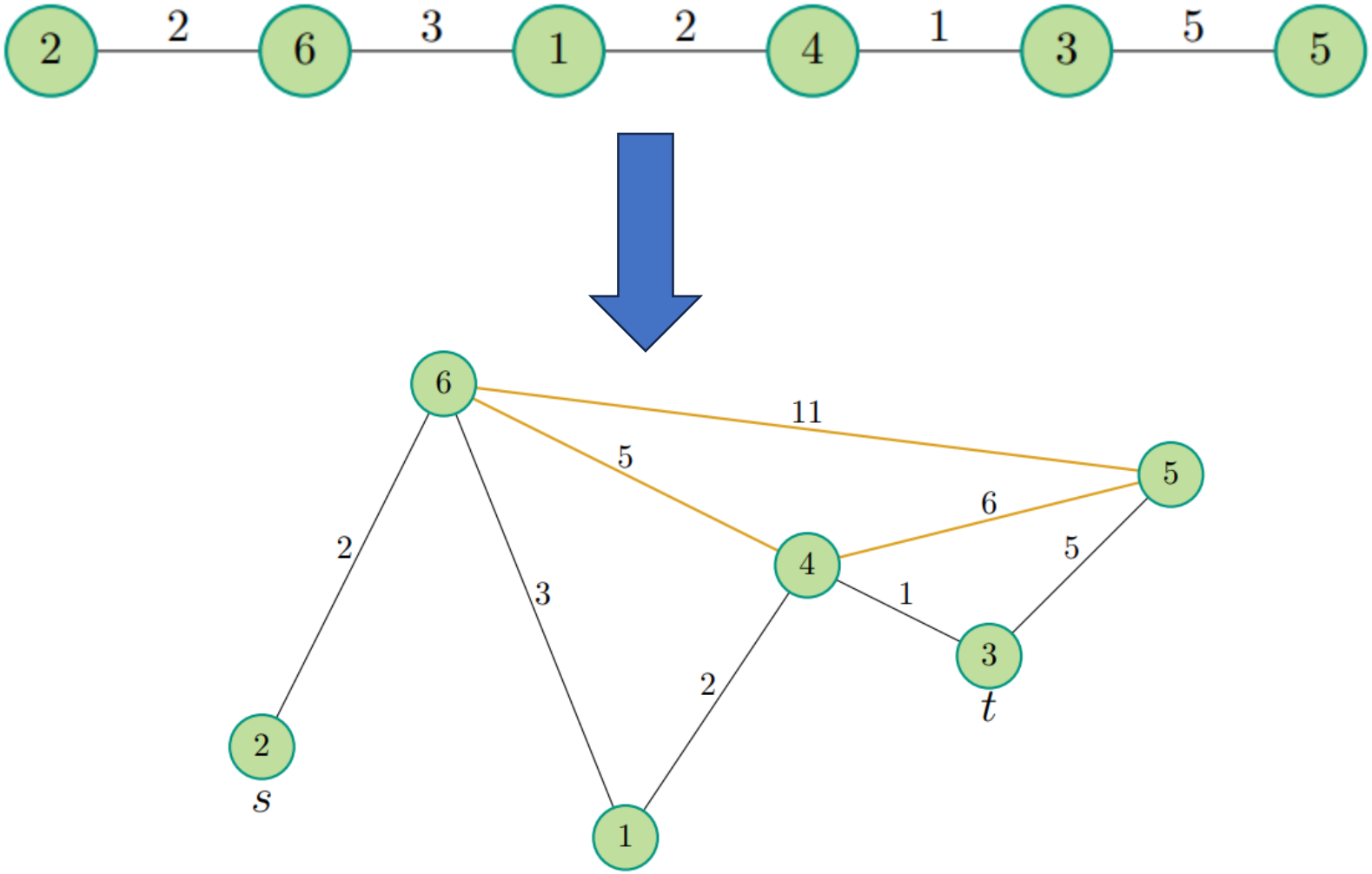
Preprocessing: Constructing an Overlay Graph

- Give each node in the input graph a unique rank according to its importance (many choices possible, but let it be arbitrary in the following)
- Assume $v_1, \ldots, v_n$ are sorted according to their rank, i.e.,
$$rank(v_1) \leq rank(v_2) \leq \ldots \leq rank(v_n)$$
- For each node v_i: contract(v_i)
- Contraction of a node v:

foreach *pair* (u, v) *and* (v, w) *of edges* **do**
  if (u, v, w) *is a unique shortest path* **then**
    add shortcut (u, w) with weight $w(u, v, w)$

Example contraction on blackboard

Larger example



Example taken from https://i11www.iti.kit.edu/_media/teaching/winter2022/algorithmengineeringpraktikum/einfuehrung.pdf

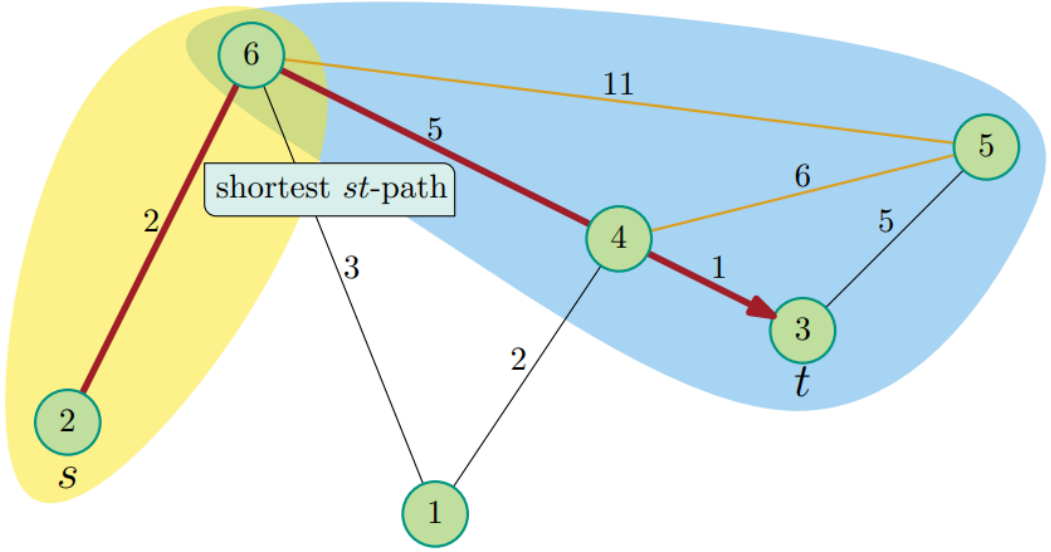
The catch

- Why would that ever be better?
  - We just added edges!
- Catch is in the query:
  - Bidirectional Dijkstra
  - From start, **only going up in rank**
  - From target, **only going up in rank**
  - **Termination:** both priority queues have priority larger than current best (s, t) path.

Algorithm 1: Query(s,t)

```
1  d↑:= ⟨∞,...,∞⟩; d↓:= ⟨∞,...,∞⟩; d:= ∞;        // tentative distances
2  d↑[s]:= 0; d↓[t]:= 0;          // start bidirectional search from s and t
3  Q↑ = {(0,s)}; Q↓ = {(0,t)}; r:= ↑;                  // priority queues
4  while (Q↑ ≠ ∅ or Q↓ ≠ ∅) and (d > min{min Q↑, min Q↓}) do
5      if Q_¬r ≠ ∅ then r:= ¬r;        // interleave direction, ¬↑=↓ and
         ¬↓=↑
6      (·,u):= Q_r.deleteMin();                                  // settle u
7      d:= min{d, d↑[u] + d↓[u]};              // u is potential candidate
8      foreach e = (u,v) ∈ E* do                                // relax edges
9          if r(e) and (d_r[u] + c(e) < d_r[v]) then      // shorter path found
10             d_r[v]:= d_r[u] + c(e);              // update tentative distance
11             Q_r.update(d_r[v],v);                  // update priority queue
12 return d;
```

THEOREM 3.1. *Algorithm 1, applied to a CH, returns the correct shortest path distance.*



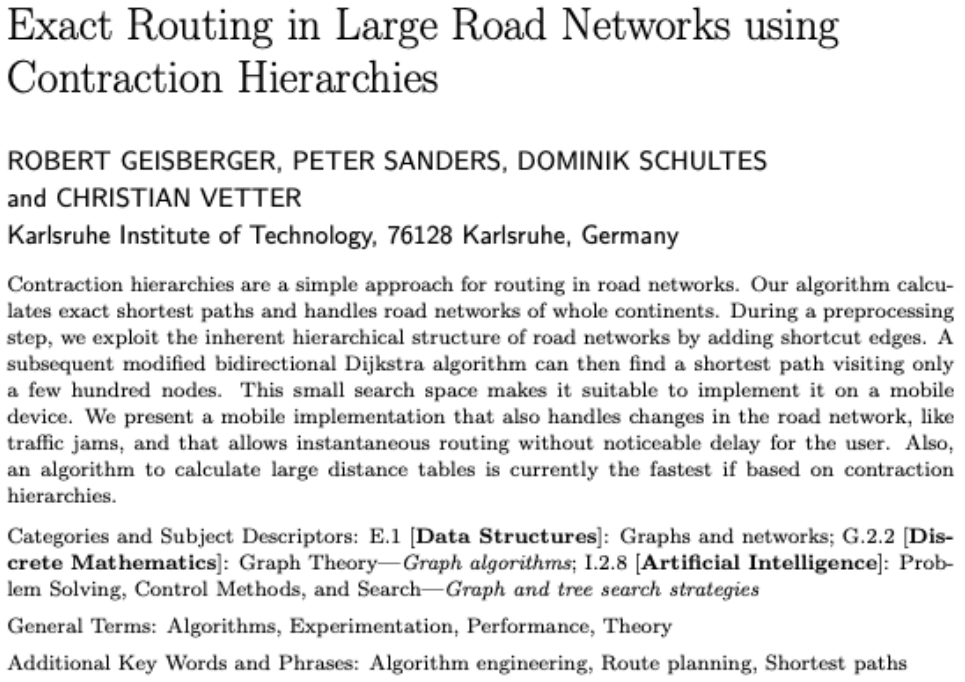
Why does this work?

- See Blackboard

Mandatory Assignment 3

- Implement contraction hierarchies and compare to bidirectional Dijkstra
- Much more open ended than other two assignments
- In groups of 3
- Due date Nov 12, 11:59

- Please follow code template on learnIT to have a good start



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ROBERT GEISBERGER, PETER SANDERS, DOMINIK SCHULTES
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Paper on learnIT

Speed-up considerations

Contraction Hierarchies works for arbitrary ordering choices, but might be really slow during preprocessing!

- How to find contraction ordering?
  - Random?
  - Edge difference
    - How many edges are added/removed in contraction
    - Want to add as few as possible
- How to change the graph?
  - Expensive: Removing edges
  - Can we ignore edges?

- How to find out if (u, v, w) is unique, i.e., (u, w) has to be added a shortcut
  - Dijkstra from u to w ignoring v?
  - When should we stop?
  - How can we initialize a search efficiently? (Don't reset all n dist values!)
  - Fixed bound on number of settled nodes?
    - Might add "unnecessary shortcuts"

A real algorithmic beauty

- Improves a well-known algorithm
- Implementation not too difficult
- Fast running times on ”instances of practical relevance”
- Correctness not obvious